

Module-1: Introduction to Statistics

* Find Mean, Mode, Median of following freq. distribution

C.I	0-10	10-20	20-30	30-40	40-50	50-60	60-70
frequency:	5	8	7	12	28	20	10

1) Mean

$$\bar{x} = \frac{\sum fx}{\sum f}$$

x = m. point of Interval

Class	x	f	fx
0-10	5	5	25
10-20	15	8	120
20-30	25	7	175
30-40	35	12	420
40-50	45	28	1,260
50-60	55	20	1,100
60-70	65	10	650

$$\bar{x} = \frac{3750}{90}$$

$$\bar{x} = 41.66$$

$$\sum f = 90 \quad \sum fx = 3750$$

2) Median:-

$$\text{Median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

l = lower limit of median class

f = frequency (freq. conesp. to m. class)

h = magnitude (or h)

N = sum of all f ($\sum f$)

c = c.f. of preceding class

Class	x	f	fx	cf
0-10	5	5	25	5
10-20	15	8	125	13
20-30	25	7	175	20
30-40	35	12	420	32
40-50	45	28	1260	60
50-60	55	20	1100	80
60-70	65	10	650	90

c

→ median class.

$$N = \sum f = 90 \quad \sum fx = 3750.$$

$$\frac{N}{2} = \frac{90}{2} = 45 \quad \text{check in c.f. next is 60.}$$

$$\text{Median} = 40 + \frac{10}{28} (45 - 32)$$

$$= 40 + \left(\frac{10}{28} \times 13 \right)$$

$$= 40 + \left(\frac{130}{28} \right)$$

$$= 40 + 4.64$$

$$= 44.64 //$$

3) Mode:-

$$\text{mode} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

l = lower limit,

h = magnitude

f_1 = frequency of modal class

f_0, f_2 = freq. of preceding & succeeding modal class.

class	x	f	fx	cf
0-10	5	5	25	5
10-20	10 ⁵	8	125	13
20-30	25	7	175	20
30-40	35	12 (6)	420	32
40-50	45	(28) (f ₁)	1260	60
50-60	55	20 (f ₂)	1100	80
60-70	65	10	650	90

$$\begin{aligned}
 \text{Mode} &= 40 + \frac{10(28-12)}{2 \times 28 - 12 - 20} \\
 &= 40 + \frac{10 \times 16}{56 - 32} \\
 &= 40 + \frac{160}{24} \\
 &= 40 + 6.66 \\
 &= 46.66 //
 \end{aligned}$$

→ Modal class.

Exampler

An incomplete frequency distribution is given as follows:

variable	frequency
10-20	12
20-30	30
30-40	? f ₁
40-50	65
50-60	? f ₂
60-70	25
70-80	18
Total	229

Given that median value is 46.
Determine missing frequencies using median formula.

Solr

$$\text{Median} = L + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

$$46 = 40 + \frac{10}{65} \left(\frac{229}{2} - (42 + f_1) \right)$$

$$46 = 40 + \frac{10}{65} \left(\frac{229}{2} - 42 - f_1 \right)$$

$$46 = 40 + \frac{10}{65} \left(\frac{229 - 84 - 2f_1}{2} \right)$$

$$46 = 40 + \frac{10}{65} \left(\frac{145 - 2f_1}{2} \right)$$

$$6 = \frac{10}{65} (72.5 - f_1)$$

$$6 = \frac{10(72.5 - f_1)}{65}$$

$$390 = 725 - 10f_1$$

$$10f_1 = 335$$

$$f_1 = 33.5 //$$

$$f_1 + f_2 = 79$$

$$f_2 = 45.5 //$$

$$N = 150 + f_1 + f_2$$

$$\text{Given } N = 229$$

$$150 + f_1 + f_2 = 229$$

$$f_1 + f_2 = 79$$

* Range Measures of Dispersion

- 1) Range
- 2) Quartile deviation
- 3) Mean deviation
- 4) Standard deviation.

1) Range

Difference b/w 2 extreme observations.

2) Quartile deviation

$$Q_D = \frac{1}{2} (Q_3 - Q_1)$$

$$Q_i = l + \frac{h}{f} \left(\frac{iN}{4} - c \right) \quad \text{where } i = 1, 2, 3$$

$$Q_1 = l + \frac{h}{f} \left(\frac{N}{4} - c \right)$$

$$Q_2 = l + \frac{h}{f} \left(\frac{N}{2} - c \right) = \text{Median}$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3N}{4} - c \right)$$

$$\text{then } Q_D = \frac{Q_3 - Q_1}{2} //$$

3) Mean Deviation about mean.

$$\text{Mean deviation} = \frac{\sum f|x - \bar{x}|}{N}$$

Mean value deviated from $\bar{x}$.

4) Standard Deviation

$$S.D. = \sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}}$$

$$\text{Square of S.D} = \text{Variance} = \sigma^2 = \frac{\sum f(x - \bar{x})^2}{N}$$

* Karl-pearson Relationr

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

* Calculate Mean deviation about ^{mean} median and coefficient of mean deviation about median for following data.

class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
frequency	18	16	15	12	10	5	2	2

solⁿ mean deviation about median $M.D = \frac{\sum f|x-\bar{x}|}{N}$ $M.D = \frac{\sum f|x-M|}{N}$

Class	freq	x	fx	x- $\bar{x}$	f x- $\bar{x}$	Cf ^{for median}
0-10	18	5	90	21.62	389.16	18
10-20	16	15	240	11.62	185.92	34
20-30	15	25	375	1.62	24.3	49
30-40	12	35	420	9.62	115.44	84
40-50	10	45	450	19.62	196.2	129
50-60	5	55	275	29.62	148.1	184
60-70	2	65	130	39.62	79.24	249
70-80	2	75	150	49.62	99.24	324

$\bar{x} = \frac{\sum fx}{N(\sum f)}$
 $= \frac{2130}{80}$
 $= 26.625$

Median class

$\checkmark N = \sum f = 80$ $\sum fx = 2130$ $\sum f|x-\bar{x}| = 1,237.24$

$M.D = \frac{\sum f|x-\bar{x}|}{N}$

Coeff

Coeff of M.D about mean = $\frac{MD(\text{mean})}{\bar{x}}$

Coeff of M.D about median = $\frac{MD(\text{median})}{\text{Median}}$

Median = $1 + \frac{h}{f} \left(\frac{N}{2} - C \right)$
 $= 20 + \frac{10}{15} (40 - 34)$
 $= 20 + \frac{2}{3} \times 6^2$
 $= 24 //$

Mean ($\bar{x}$) = $\frac{\sum fx}{\sum f}$
 $= \frac{2130}{80}$
 $= 26.625$

m.D about median

class	x	f	Cf	$ x - M $	$f x - M $
0-10	5	18	18	19	342
10-20	15	16	34	9	144
20-30	25	15	49	1	15
30-40	35	12	61	11	132
40-50	45	10	71	21	210
50-60	55	5	76	31	155
60-70	65	2	78	41	82
70-80	75	2	80	51	102

$$\sum f = 80 \quad \sum f = 80 = N$$

$$\sum f|x - M| = 1182$$

$$\text{Mean deviation about Median} = \frac{\sum f|x - M|}{N} = \frac{1182}{80} = 14.775$$

$$\text{Coefficient of m.D about Median} = \frac{14.775}{\text{median}} = \frac{14.775}{24} = 0.615 //$$

m.D about mean

class	x	f	fx	$ x - \bar{x} $	$f x - \bar{x} $
0-10	5	18	90	21.625	389.25
10-20	15	16	240	11.625	186
20-30	25	15	375	1.625	24.375
30-40	35	12	420	8.375	100.5
40-50	45	10	450	18.375	183.75
50-60	55	5	275	28.375	141.875
60-70	65	2	130	38.375	76.75
70-80	75	2	150	48.375	96.75

$$\sum f = 80 \quad \sum fx = 2130$$

$$\sum f|x - \bar{x}| = 1199.25 \approx 1200$$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{2130}{80} = 26.625 //$$

$$\text{Mean deviation about Mean} = \frac{\sum f|x - \bar{x}|}{N} = \frac{1200}{80} = 15 //$$

$$\text{Coefficient of m.D about Mean} = \frac{15}{\bar{x}} = \frac{15}{26.625} = 0.563$$

* Coefficient of Range $\frac{L-S}{L+S}$

* Range $L-S$

Ex: 35, 40, 52, 29, 51, 46, 27, 30, 30, 23.

$$\text{Range} = L - S = 52 - 23 = 29 //$$

$$\text{Coeff. of Range} = \frac{L-S}{L+S} = \frac{52-23}{52+23} = \frac{29}{75} = 0.386 //$$

N	35-45	45-55	55-65	65-75	75-85
	40	50	60	70	80

* Quartile Deviation

$$Q_1 = l + \frac{h}{f} \left[\frac{iN}{4} - c \right]$$

$$Q_D = \frac{Q_3 - Q_1}{2}$$

$$\text{Coefficient of Q.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Prob Find the Quartile deviation & quartile coefficient of dispersion for following data.

Solⁿ

class	frequency	x	cf
0-10	8	5	8
10-20	20	15	28
20-30	34	25	62 $\leftarrow Q_1$
30-40	46	35	108
40-50	28	45	136 $\leftarrow Q_3$
50-60	14	55	150
60-70	10	65	160

$$N = \sum f = 160$$

$$\frac{N}{2} = \frac{160}{2} = 80$$

$$Q_D = \frac{Q_3 - Q_1}{2}$$

$$\frac{N}{4} = \frac{160}{4} = 40$$

$$Q_1 = l + \frac{h}{f} \left[\frac{iN}{4} - c \right]$$

$$= 20 + \frac{10}{34} [40 - 28]$$

$$= 20 + \frac{10}{34} [12]$$

$$= 23.529 //$$

$$Q_3 = l + \frac{h}{f} \left[\frac{3N}{4} - c \right]$$

$$\frac{3N}{4} \Rightarrow 3 \times 40 = 120$$

$$= 40 + \frac{10}{28} [120 - 108]$$

$$= 40 + \frac{10}{28} [12]$$

$$= 44.285 //$$

$$Q_D = \frac{Q_3 - Q_1}{2} = \frac{44.285 - 23.529}{2} = \frac{20.756}{2} = 10.378 //$$

$$\text{Coeff. of Q.D} = \frac{44.285 - 23.529}{44.285 + 23.529} = \frac{20.756}{67.814} = 0.306 //$$

* Standard Deviation:-

$$S.D = \sigma = \sqrt{\frac{\sum f(x-\bar{x})^2}{N}}$$

$$\sum f = N$$

$$\text{Variance} = \sigma^2 = \frac{\sum f(x-\bar{x})^2}{N}$$

prob:- Find standard deviation & coefficient of standard variation for following data.

Class	freq	x	fx	$(x-\bar{x})^2$	$f(x-\bar{x})^2$
0-10	8	5	40	855.5625	6844.5
10-20	20	15	300	370.5625	7411.25
20-30	34	25	850	85.5625	2909.125
30-40	46	35	1610	0.5625	25.875
40-50	28	45	1260	115.5625	3235.75
50-60	14	55	770	430.5625	6027.875
60-70	10	65	650	945.5625	9455.625
$\sum f = 160$		$\sum fx = 5480$		$\sum f(x-\bar{x})^2 = 35910$	

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{5480}{160} = 34.25 //$$

$$\sigma = S.D = \sqrt{\frac{35910}{160}} = \sqrt{224.4375} = 14.98$$

Coefficient of Variation (Variability) $\Rightarrow$ $C.V = 100 \times \frac{\sigma}{\bar{x}}$

$$C.V = 100 \times \frac{14.98}{34.25}$$

$$= 100 \times 0.437$$

$$= 43.737 //$$